



Calculating theoretical probabilities from Venn diagrams (two events)

Task A: Theoretical probabilities of combined events

1) Use the Venn diagram to find the probabilities.

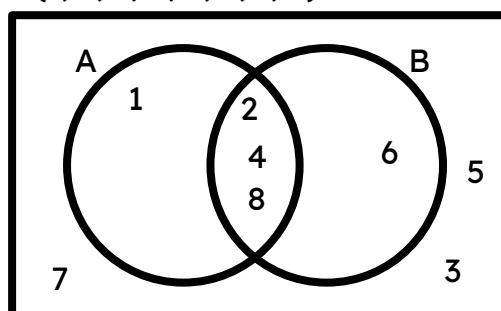
$$a) P(A \text{ and } B) = \frac{3}{8} = 37.5\% = 0.375$$

$$b) P(A \text{ and not } B) = \frac{1}{8} = 12.5\% = 0.125$$

$$c) P(B \text{ and not } A) = \frac{1}{8} = 12.5\% = 0.125$$

$$d) P(\text{not } A \text{ and not } B) = \frac{3}{8} = 37.5\% = 0.375$$

$$\xi = \{1, 2, 3, 4, 5, 6, 7, 8\}$$



2) Use the Venn diagram to write the probabilities.

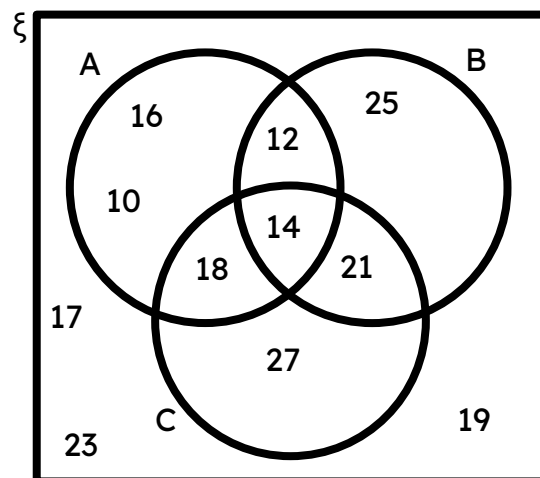
$$a) P(A \text{ and } C) = \frac{2}{11}$$

$$b) P(B \text{ and not } C) = \frac{2}{11}$$

$$c) P(\text{not } C \text{ and not } A) = \frac{4}{11}$$

$$d) P(A \text{ and not } B) = \frac{3}{11}$$

$$e) P(\text{not } A \text{ and } B) = \frac{2}{11}$$



3) This Venn diagram shows the number of outcomes in each event.

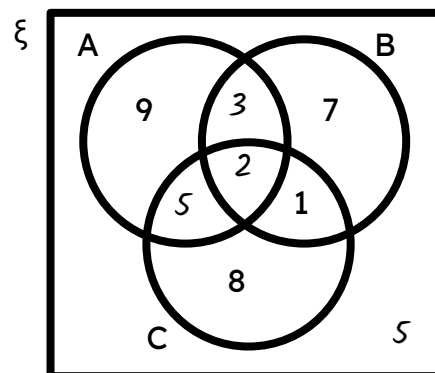
There are 40 outcomes in the sample space.

Complete the Venn diagram given these probabilities.

$$P(\text{not } A) = \frac{21}{40}$$

$$P(A \text{ and not } C) = \frac{3}{10}$$

$$P(B) = \frac{13}{40}$$



Task B: Further combinations of events

1) Use the Venn diagram to answer the following questions.

a) List the outcomes that are in event A.

$\{12, 15, 18, 20\}$

b) List the outcomes that are not in event B.

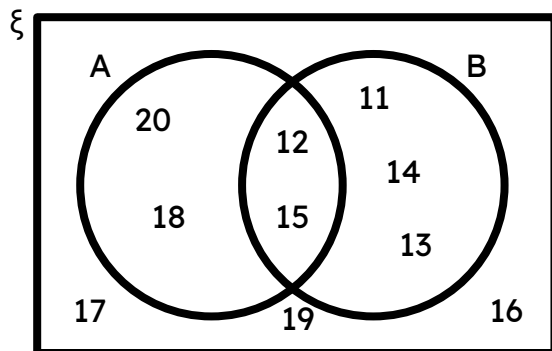
$\{16, 17, 18, 19, 20\}$

c) Hence, list the outcomes that are in event A or not in event B.

$\{12, 15, 16, 17, 18, 19, 20\}$

d) Find $P(A \text{ or not } B)$

$$= \frac{7}{10} = 0.7 = 70\%$$



2) Use the Venn diagram to answer the following questions.

a) Find $P(A \text{ or } B)$

$$= \frac{8}{10} = \frac{4}{5} = 0.8 = 80\%$$

b) Find $P(A \text{ or not } B)$

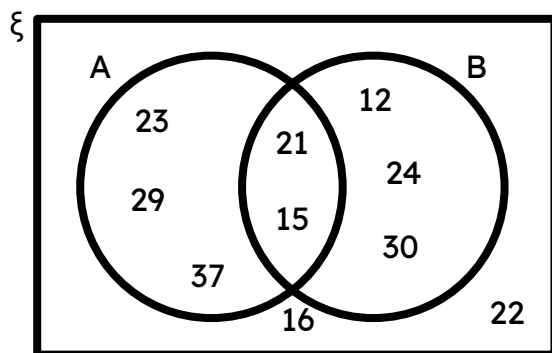
$$= \frac{7}{10} = 0.7 = 70\%$$

c) Find $P(\text{not } A \text{ or not } B)$

$$= \frac{8}{10} = \frac{4}{5} = 0.8 = 80\%$$

d) Find $P(\text{only } A \text{ or only } B)$

$$= \frac{6}{10} = \frac{3}{5} = 0.6 = 60\%$$



3) Use the Venn diagram to answer the following questions.

a) Find $P(A \text{ or } B)$

$$= \frac{7}{10} = 0.7 = 70\%$$

b) Find $P(\text{not } A \text{ or } C)$

$$= \frac{8}{10} = \frac{4}{5} = 0.8 = 80\%$$

c) Find $P(\text{not } A \text{ or not } B)$

$$= \frac{8}{10} = \frac{4}{5} = 0.8 = 80\%$$

d) Find $P(\text{only } C \text{ or not } B)$

$$= \frac{5}{10} = \frac{1}{2} = 0.5 = 50\%$$

